

Holoscan Sensor Bridge Bring-up Guide

NVIDIA AGX Thor + Agilex 5 Modular Development Kit

This guide summarizes the customer bring-up flow used for the Altera/NVIDIA Holoscan Sensor Bridge demo on NVIDIA AGX Thor with Agilex 5 Modular Development Kits. It covers the working 10GbE and 25GbE paths, the issues encountered, and the fixes applied.

1. Reference Documents

- Jetson AGX Thor ISO installation:
<https://docs.nvidia.com/jetson/agx-thor-devkit/user-guide/latest/quick_start.html#>
 - Holoscan Sensor Bridge host setup, AGX Thor tab:
<<https://docs.nvidia.com/holoscan/sensor-bridge/2.5.0/setup.html#sd-tab-item-3>>
 - Altera HSB 25GbE reference design:
<https://github.com/altera-fpga/holoscan-sensor-bridge/tree/altera-release-2.6.0/fpga/altera/AGX_5E_065A_Modular_DevKit_HSB_MIPi_25GbE>
 - Altera HSB 10GbE reference design:
<https://github.com/altera-fpga/holoscan-sensor-bridge/tree/altera-release-2.6.0/fpga/altera/AGX_5E_065B_Modular_DevKit_HSB_MIPi_10GbE>
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2. System Used

Host

- NVIDIA Jetson AGX Thor Developer Kit
- JetPack 7.2 / L4T R39.2

Verify:

```
cat /etc/nv_tegra_release
```

What this does / why required:

- Confirms that the Thor is running the expected JetPack/L4T release.
- The 10G/25G switching capsule procedure used in this guide is for JetPack 7.2 / L4T R39.2.

Expected:

```
# R39 (release), REVISION: 2.0
```

FPGA Images

10GbE / Group B:

```
AGX_5E_065B_Modular_DevKit_HSB_MIPi_10GbE.sof
```

25GbE / Group A:

```
AGX_5E_065A_Modular_DevKit_HSB_MIPi_25GbE.sof
```

3. AGX Thor Base Installation

Follow the Jetson AGX Thor ISO quick start guide.

High-level flow:

1. Download the Jetson AGX Thor ISO image.
2. Create the bootable USB installer.
3. Boot AGX Thor from the USB installer.
4. Install the BSP to NVMe.
5. Remove the USB installer.
6. Boot from NVMe and complete Ubuntu OEM setup.
7. Verify JetPack/L4T version.

Reference:

https://docs.nvidia.com/jetson/agx-thor-devkit/user-guide/latest/quick_start.html#

4. Host Setup for Holoscan Sensor Bridge

Use the AGX Thor tab in the Holoscan Sensor Bridge host setup guide.

Reference:

<https://docs.nvidia.com/holoscan/sensor-bridge/2.5.0/setup.html#sd-tab-item-3>

Clone the Altera release branch:

```
git clone -b altera-release-2.6.0 https://github.com/altera-fpga/holoscan-sensor-bridge.git
cd holoscan-sensor-bridge
git lfs pull
```

What this does / why required:

- git clone -b altera-release-2.6.0 gets the Altera-specific Holoscan Sensor Bridge release with the AGX5 example applications.
- git lfs pull downloads large model/data/binary assets tracked by Git LFS.

Build the demo container:

```
sh docker/build.sh --igpu
```

What this does / why required:

- Builds the Holoscan Sensor Bridge demo image for AGX Thor's integrated GPU path.
- The first build can take significant time because it downloads base images, installs dependencies, and builds the Hololink software.

Start the demo container:

```
xhost +
sh docker/demo.sh
```

What this does / why required:

- xhost + allows GUI applications in the container to connect to the Thor desktop display.

- sh docker/demo.sh starts the prebuilt demo container with host networking, GPU access, device access, and repository mounts.

Inside the container, use:

```
hololink-enumerate
```

What this does / why required:

- Discovers Holoscan Sensor Bridge devices on the configured network.
- A successful enumerate proves the Ethernet link, IP route, and FPGA HSB endpoint are working before camera demos are started.

The tool is installed under /usr/local/bin, so ./tools/enumerate/hololink-enumerate may not exist in the container.

5. Common Network Configuration

The FPGA/HSB default IP is:

```
192.168.0.2
```

Configure Thor:

```
EN0=mgbe0_0
```

```
sudo nmcli connection delete hololink-$EN0 2>/dev/null || true
sudo nmcli con add con-name hololink-$EN0 ifname $EN0 type ethernet ip4 192.168.0.101/24
sudo nmcli connection modify hololink-$EN0 +ipv4.routes 192.168.0.2/32
sudo nmcli connection up hololink-$EN0
```

What this does / why required:

- EN0=mgbe0_0 selects the Thor MGBE port connected to the FPGA.
- nmcli connection delete removes any stale Hololink profile so the IP setup is clean.
- nmcli con add assigns Thor the static host IP 192.168.0.101/24.
- +ipv4.routes 192.168.0.2/32 adds a direct route to the FPGA/HSB IP.
- connection up activates the interface profile.

Test:

```
ping 192.168.0.2
```

What this does / why required:

- Confirms basic IP reachability to the FPGA.
- If this fails with Destination Host Unreachable, debug link/cable/FPGA programming before running Holoscan applications.

Check link status:

```
for i in 0 1 2 3; do
  echo "--- mgbe${i}_0 ---"
  sudo ethtool mgbe${i}_0 | grep -E "Speed|Link detected"
done
```

What this does / why required:

- Reports which Thor MGBE port has carrier and at what speed.
- For 10G, expect Speed: 10000Mb/s and Link detected: yes.

- For 25G, expect Speed: 25000Mb/s and Link detected: yes.

6. 10GbE Demo Flow

Program the 10GbE FPGA image:

```
AGX_5E_065B_Modular_DevKit_HSB_MIIPI_10GbE.sof
```

Expected Thor link:

```
mgbe0_0
Speed: 10000Mb/s
Link detected: yes
```

Successful HSB enumeration example:

```
mac_id=CA:FE:C0:FF:EE:10
hsb_ip_version=0x2603
ip_address=192.168.0.2
fpga_uuid=7b1fa8c7-31aa-44b6-abcc-eac134461fdc
interface=mgbe0_0
```

Run camera test:

```
python3 examples/linux_agx5_player.py --cam 0 --lines 1080 --frame-rate 30
```

What this does / why required:

- Starts a visible single-camera pipeline using camera 0.
- 1080p30 is a conservative first demo setting and is easier to debug than starting at maximum resolution/frame rate.

Headless validation:

```
python3 examples/linux_agx5_player.py --cam 0 --lines 1080 --frame-rate 30 --headless --frame-limit 100
```

What this does / why required:

- Runs the camera pipeline without opening a display window.
- --frame-limit 100 exits automatically after 100 frames, which is useful for quick validation or remote debugging.

Stereo:

```
python3 examples/linux_agx5_player_stereo.py
```

What this does / why required:

- Runs both physical cameras in the stereo example after each camera works individually.
-

7. 25GbE Demo Flow

Important

AGX Thor does not support runtime switching between 10GbE x4 and 25GbE x4 through nmcli or normal Linux network settings. The Thor boot image / QSPI boot slot must be switched.

Capsule files used:

Tegra_Thor_BL_7.2_10G.Cap
Tegra_Thor_BL_7.2_25G.Cap

Current slot mapping from this bring-up:

slot 0 = 25G
slot 1 = 10G

Switch to 25G:

```
sudo nvbootctrl set-active-boot-slot 0  
sudo reboot
```

What this does / why required:

- Selects the Thor boot slot that contains the 25GbE-capable bootloader/QSPI configuration.
- A reboot is required because AGX Thor does not switch MGBE 10G/25G mode at runtime.

Switch to 10G:

```
sudo nvbootctrl set-active-boot-slot 1  
sudo reboot
```

What this does / why required:

- Selects the Thor boot slot that contains the 10GbE configuration.
- Use this to return to the stable 10G demo path.

Check active slot and speed:

```
sudo nvbootctrl get-current-slot  
cat /sys/devices/platform/bus@0/*/net/mgbe*/speed
```

What this does / why required:

- nvbootctrl get-current-slot shows which boot slot is active.
- The speed files confirm whether the MGBE ports are configured as 10G or 25G after reboot.

Expected for 25G:

```
0  
25000  
25000  
25000  
25000
```

Program the 25GbE FPGA image:

AGX_5E_065A_Modular_DevKit_HSB_MIPI_25GbE.sof

Expected Thor link:

```
mgbe0_0  
Speed: 25000Mb/s  
Link detected: yes
```

Run:

```
hololink-enumerate  
python3 examples/linux_agx5_player.py --cam 0 --lines 1080 --frame-rate 30
```

What this does / why required:

- First confirms that the FPGA enumerates in 25G mode.
- Then runs a basic camera pipeline before attempting multi-camera or AI demos.

For two physical cameras:

```
python3 examples/agx5_multiviewer.py --cam 2 --lines 2160 --frame-rate 30 --receiver-type coe
```

What this does / why required:

- Runs the Group A 25G multiviewer with two real cameras.
- --receiver-type coe uses Thor's Camera-over-Ethernet path, which is the recommended 25G demo path.
- Use --cam 2 when only two physical cameras are installed. Do not use --cam 8 unless the demo scenario requires replicated streams and the platform has been validated for it.

25G CoE Stability Workaround on AGX Thor

For the 25G demo on AGX Thor, use CoE mode demos only. If running the demo repeatedly, refresh the MGBE interface before every run after the first one:

```
sudo ip link set mgbe0_0 down
sudo ip link set mgbe0_0 up
```

What this does / why required:

- Briefly resets the Thor MGBE interface state between repeated 25G CoE demo runs.
- This avoids repeated-run stability issues observed on AGX Thor with 25G CoE demos.

Equivalent one-line command:

```
sudo ip link set mgbe0_0 down ; sudo ip link set mgbe0_0 up
```

This workaround was recommended for AGX Thor 25GbE demos to avoid issues seen on repeated CoE runs.

For debugging only, if CoE remains unstable, test with Linux receiver:

```
python3 examples/agx5_multiviewer.py --cam 2 --lines 1080 --frame-rate 30 --receiver-type linux
```

What this does / why required:

- Forces the Linux socket receiver instead of CoE.
- This is useful as a diagnostic fallback, but CoE mode is the intended 25G demo path on Thor.

8. MAX10 MIPI Power Fix

Symptom

Holoscan applications reached the FPGA/HSB, but camera configuration failed with I2C transaction errors:

```
hololink._hololink.TransactionError: i2c_transaction i2c_address=0x37
hololink._hololink.TransactionError: i2c_transaction i2c_address=0x1a
```

Root Cause

Early production MDK boards, both Group A and Group B, shipped with a bug in the MAX10 image. The buggy MAX10 image does not apply power to the MIPI connectors. Without MIPI connector power, camera I2C communication is impossible.

Customer release production boards are expected to ship with a fixed MAX10 image and should be plug-and-play. Currently available early boards may need MAX10 reflashing.

Fix Applied

1. Switch MAX10 onto the JTAG chain by setting SW4 on the carrier board to ON. This is the single switch nearest the HDMI connector.
2. Use Quartus Programmer GUI to reflash MAX10 with:

```
max10_top_rtl_v1p1p6_fw_v2p0p1.pof
```

3. Power off the board.
4. Set SW4 back to OFF so the FPGA appears on the JTAG chain again.

After this MAX10 update, camera I2C configuration succeeded.

Additional Note

Initial production MDK batches may have a carrier micro USB hub that is susceptible to ESD. If the carrier micro USB is not detected, for example Windows fails to read descriptors, use the standard JTAG connector and a Byte Blaster.

9. AI Demo Examples

YOLOv8 Body Pose

Inside container:

```
apt-get update && apt-get install -y ffmpeg
pip3 install ultralytics onnx

cd examples
yolo export model=yolov8n-pose.pt format=onnx
trtexec --onnx=yolov8n-pose.onnx --saveEngine=yolov8n-pose.engine.fp32
cd -
```

What this does / why required:

- Installs temporary model conversion dependencies inside the container.
- Exports the YOLOv8 pose model to ONNX.
- Builds a TensorRT engine so the body-pose demo can run efficiently on Thor.

Run:

```
python3 examples/linux_body_pose_estimation_agx5.py --cam 0 --lines 1080 --frame-rate 30
```

What this does / why required:

- Runs live camera capture plus YOLOv8 body-pose inference and overlay visualization.

TAO PeopleNet

Download model:

```
curl -L 'https://api.ngc.nvidia.com/v2/models/org/nvidia/team/tao/peoplenet/pruned_quantized_decrypted_v2.3.3/files?redirect=true&path=resnet34_peoplenet_int8.onnx' -o examples/resnet34_peoplenet_int8.onnx
```

What this does / why required:

- Downloads the TAO PeopleNet ONNX model into the examples directory where the AGX5 script expects it.
- curl -L follows the NGC redirect URL.

Run:

```
python3 examples/linux_tao_peoplenet_agx5.py --cam 0 --lines 1080 --frame-rate 30
```

What this does / why required:

- Runs live camera capture plus PeopleNet inference.
- The first run builds and caches a TensorRT engine, so it can take several minutes before the window appears.

First run builds a TensorRT engine and can take several minutes.

If the PeopleNet AGX5 example crashes in InferenceOp, disable metadata in examples/linux_tao_peoplenet_agx5.py:

```
self.is_metadata_enabled = False
```

10. Display and Runtime Notes

If Holoviz/GLFW fails:

```
Failed to initialize glfw
Authorization required, but no authorization protocol specified
XDG_RUNTIME_DIR is invalid or not set
```

Run on Thor host before starting Docker:

```
xhost +local:root
xhost +local:docker
xhost +
```

What this does / why required:

- Grants the root user/container permission to open GUI windows on the Thor desktop.
- Required for Holoviz display windows.

Inside Docker:

```
export XDG_RUNTIME_DIR=/tmp/runtime-root
mkdir -p $XDG_RUNTIME_DIR
chmod 700 $XDG_RUNTIME_DIR
```

What this does / why required:

- Creates a valid runtime directory for GUI/Vulkan/GLFW components inside the container.
- Prevents XDG_RUNTIME_DIR is invalid or not set errors.

For receiver buffer warnings:

```
echo 10420224 | sudo tee /proc/sys/net/core/rmem_max
```

What this does / why required:

- Increases the Linux receive buffer limit.
- Helps avoid dropped packets or unreliable performance in socket-based receiver demos.

Inside the container as root, omit sudo:

```
echo 10420224 > /proc/sys/net/core/rmem_max
```

Use Ctrl+C to stop demos. Avoid Ctrl+Z, which suspends jobs and can leave resources open.

11. Known Issues and Lessons Learned

- Thor defaults to 10GbE mode.
- 25GbE FPGA images need Thor booted in 25GbE mode.
- 10G/25G mode switching requires boot slot / capsule handling, not nmcli.
- Early MDK boards may need MAX10 reflashing to power MIPI connectors.
- Camera I2C errors can be caused by missing MIPI connector power, not only cable orientation.
- hololink-enumerate confirms the FPGA/HSB network path.
- GUI failures are often X11 authorization issues.
- .sof FPGA programming is volatile; reprogram after FPGA board power cycle.
- For customer demos on AGX Thor, 10GbE and Linux socket mode are the most stable paths. 25GbE CoE is available but may show stability issues under some multi-stream scenarios.